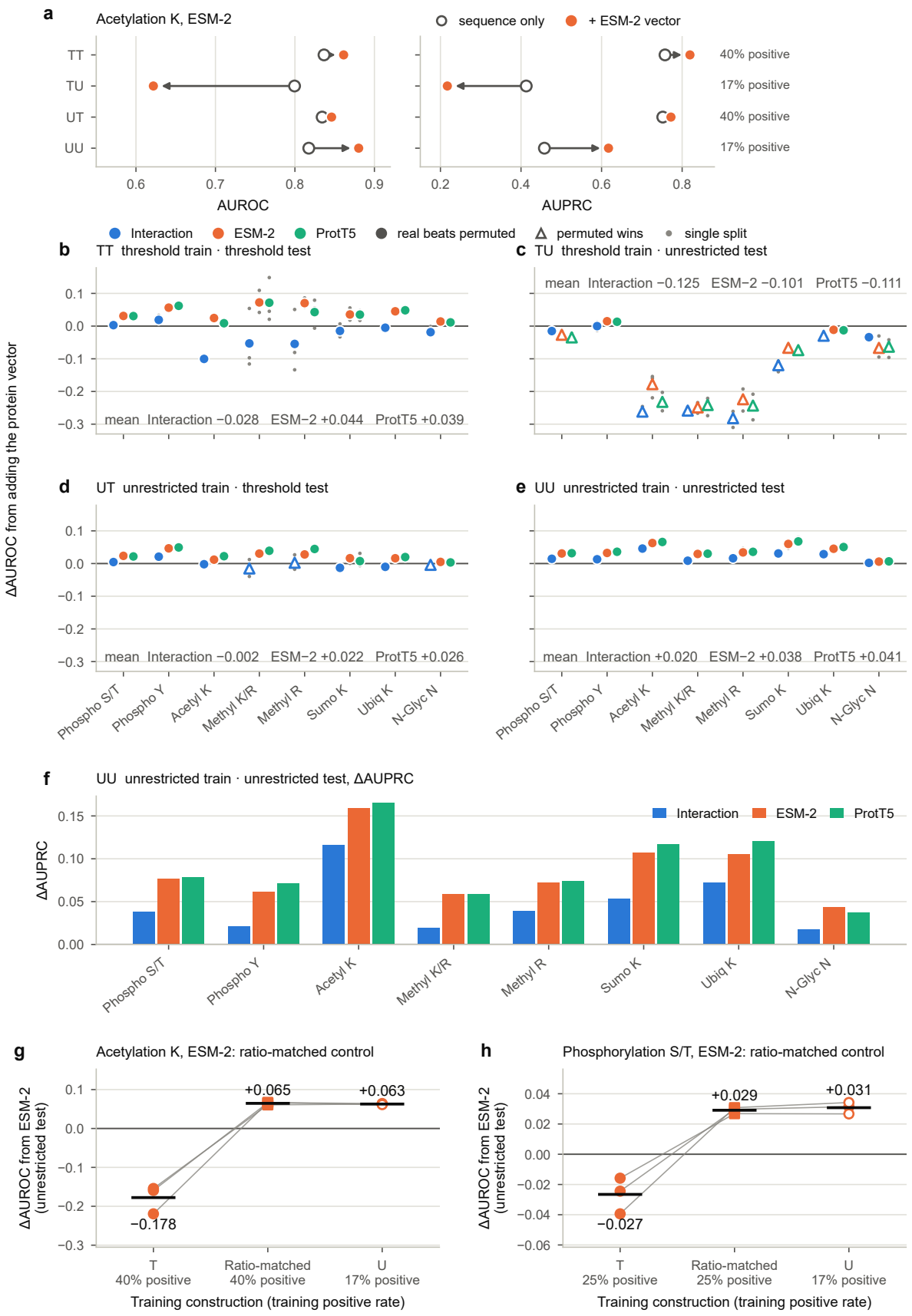

a
Acetylation K, ESM-2
sequence only
+ ESM-2 vector
TT
TU
UT
UU
40% positive
17% positive
40% positive
17% positive
0.6
0.7
0.8
0.9
AUROC
0.2
0.4
0.6
0.8
AUPRC
Interaction
ESM-2
ProtT5
real beats permuted
permuted wins
single split
b
TT threshold train · threshold test
mean Interaction −0.028 ESM−2 +0.044 ProtT5 +0.039
c
TU threshold train · unrestricted test
mean Interaction −0.125 ESM−2 −0.101 ProtT5 −0.111
ΔAUROC from adding the protein vector
d
UT unrestricted train · threshold test
mean Interaction −0.002 ESM−2 +0.022 ProtT5 +0.026
e
UU unrestricted train · unrestricted test
mean Interaction +0.020 ESM−2 +0.038 ProtT5 +0.041
Phospho S/T
Phospho Y
Acetyl K
Methyl K/R
Methyl R
Sumo K
Ubiq K
N-Glyc N
f
UU unrestricted train · unrestricted test, ΔAUPRC
ΔAUPRC
Interaction
ESM-2
ProtT5
g
Acetylation K, ESM-2: ratio-matched control
ΔAUROC from ESM-2 (unrestricted test)
+0.065
+0.063
−0.178
T
40% positive
Ratio-matched
40% positive
U
17% positive
Training construction (training positive rate)
h
Phosphorylation S/T, ESM-2: ratio-matched control
ΔAUROC from ESM-2 (unrestricted test)
+0.029
+0.031
−0.027
T
25% positive
Ratio-matched
25% positive
U
17% positive
Training construction (training positive rate)